

EBE Core Lock v1.0

The S-Dynamic Foundation

Mathematical Engine for Entropy-Bounded Empiricism

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Abstract

This document specifies EBE Core Lock v1.0, a proposed dynamical mechanism underlying Entropy-Bounded Empiricism (EBE). The model defines a running parameter $f(S)$ with constant negative beta function, uniquely selecting an exponential form. Physical observables project via accumulated phase $\Phi(S)$, yielding the proxy-independent prediction $d \ln v / d \ln c = \text{constant}$. **The primary testable signature lies in correlations between running constants rather than their individual amplitudes.** Current null results on varying constants constrain coupling coefficients but do not falsify the framework itself.

Keywords: entropy-bounded empiricism, RG-flow, emergent constants, correlation signature, falsifiable prediction

Epistemic Status: Layer B (Hypothetical). Mathematical engine compatible with current null results; predictive at correlation level.

1 Introduction and Relation to EBE

Entropy-Bounded Empiricism (EBE) (1) provides a regime-based epistemic framework where physical validity depends on entropy state (S-axis) and epistemic strength depends on assumption load (L-axis). Core Lock supplies the dynamical foundation: an RG-flow along the entropy axis.

Layer	Framework	Role
Dynamical Foundation	Core Lock	Mathematical engine (this document)
Epistemic Framework	EBE	Classification of regimes and claims
Measurement Protocol	RCMP	Rules for regime-consistent interpretation

Table 1: Hierarchical relationship between frameworks.

2 Core Lock Specification

2.1 The Grid State Function $f(S)$

The fundamental driver is the **Grid State Function** $f(S)$, describing the effective stiffness of the vacuum grid as a function of entropy state S . This is a running coupling parameter along the entropy axis—not a field in spacetime.

2.2 The Beta-Constraint

The beta function is constant:

$$\beta_f(S) \equiv \frac{d \ln f}{dS} = -\frac{1}{\Delta S} \quad (1)$$

where ΔS is the characteristic scale parameter. This constraint yields the unique solution:

$$\boxed{f(S) = f_0 \exp\left(-\frac{S}{\Delta S}\right)} \quad (2)$$

3 Conservation Law: Information Length

The **Running Phase** $\Phi(S)$ represents accumulated information length:

$$\Phi(S) = \int_0^S \frac{dS'}{f(S')} = \frac{\Delta S}{f_0} \left[\exp\left(\frac{S}{\Delta S}\right) - 1 \right] \quad (3)$$

The total information length over the physical domain is an invariant:

$$\mathcal{I} = \Phi(S_1) = \text{constant} \quad (4)$$

4 Projection Laws

Physical parameters project via Φ :

$$\ln\left(\frac{X}{X_0}\right) = a_X \cdot \Phi(S) \quad (5)$$

where a_X is the coupling coefficient for parameter X .

Critical distinction: The engine (f , Φ) is internal dynamics. Observable quantities (α , μ , etc.) are **possible proxies** requiring specific projection laws and coupling coefficients—these are **not** determined by the engine itself.

5 Falsification Criterion

5.1 The Proxy-Independent Test

Combining projections for two parameters, the hidden phase cancels:

$$\boxed{\frac{d \ln v}{d \ln c} = \frac{a_v}{a_c} = \text{constant}} \quad (6)$$

Primary signature: The test is not whether individual constants vary, but whether any detected variations show **correlated behavior with constant log-log slope**.

5.2 Empirical Constraints from Precision Cosmology

Current observational limits on varying constants place stringent constraints on coupling coefficients if α or μ are identified as direct projections of $\Phi(S)$.

Interpretation: Direct identification of α or μ as primary projections with $\mathcal{O}(1)$ coupling coefficients is incompatible with current bounds. However, **Core Lock itself is not constrained** unless a concrete projection to specific observables is specified.

Two scenarios remain viable:

Observable	Redshift	Limit	Implied $ a $ bound
$\Delta\alpha/\alpha$ (quasar)	$z \sim 3$	$< 10^{-5}$	$< 5 \times 10^{-5}$
$\Delta\mu/\mu$ (H ₂)	$z \sim 2-3$	$< 10^{-5}$	$< 5 \times 10^{-5}$
$\Delta\alpha/\alpha$ (methanol)	$z \sim 0.89$	$< 10^{-7}$	$< 10^{-6}$
$\Delta\alpha/\alpha$ (JWST)	$z \sim 3-10$	$< 10^{-4}$	$< 2 \times 10^{-4}$

Table 2: Observational constraints on varying constants and implied coupling bounds.

1. **Weak coupling:** If $|a| < 10^{-6}$, individual variations fall below detection thresholds, but the **correlation signature** (constant log-log slope between any varying pairs) remains testable.
2. **Alternative S -mapping:** If $S(z)$ grows slower than $\ln(1+z)$, observable variation concentrates at $z \gg 10$ (early universe / CMB epoch).

Bottom line: Core Lock predicts that **if** any variations in fundamental constants are detected at any level, they must exhibit correlated behavior with ratio $a_v/a_c = \text{constant}$. This correlation has not been directly tested in current literature.

6 Numerical Verification

Internal consistency simulation confirms: constant $\beta \rightarrow$ exponential $f(S)$, finite invariant $\mathcal{I}$, and exact slope stability ($d \ln v / d \ln c = a_v/a_c$) across the full S -range. This validates the mathematical structure, not empirical truth.

7 Epistemic Status and Limitations

7.1 What Core Lock Is

- ✓ Mathematically consistent internal structure
- ✓ Compatible with EBE regime architecture and RCMP methodology
- ✓ Provides falsifiable correlation signature
- ✓ Unfalsified by current null results on varying constants

7.2 What Core Lock Is Not

- × Not empirically established
- × Not a prediction of large variations at low redshift
- × Not a specification of which observables are projections of $\Phi(S)$

7.3 Open Questions

1. **Operational definition of S :** What physical quantity does S correspond to? Without this, the framework risks being “just a reparametrization.”
2. **Physical basis for coupling coefficients:** Why should $|a|$ be small? Is there a symmetry or selection principle?
3. **$S(z)$ mapping:** The relationship between entropy parameter and redshift is not derived from first principles.

8 Conclusion

EBE Core Lock v1.0 provides a mathematical engine for entropy-driven regime transitions. The framework is minimal (one scale parameter), constrained (constant β forces exponential form), and testable (correlation signature). Current null results on varying constants are consistent with weak coupling or alternative S -mappings.

The critical test is not amplitude but correlation: Any future detection of varying constants, even at 10^{-8} level, must exhibit constant log-log slope between varied quantities to be consistent with Core Lock.

References

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Version History

Version	Date	Changes
1.0	February 2026	Initial specification with empirical constraints